1 A Detailed Proof of the 15-Step Genesis Sequence

This section gives a step-by-step proof of the current 15-step PEN genesis sequence in a form intended for a journal proof spine rather than for a developer report. The global mathematical input comes from the established PEN theorems in **??????????**, while the step-local selection facts come from the verified strict trace generated by `cabal run ab-initio - -strict -max-steps 15 -skip-validation -skip-mcts`. By **??**, that strict run is the canonical source for the selection ordering, the bar clearances, and the quantitative values (ν, κ, ρ) used below.

1.1 Proof Contract

Three conventions govern the proof.

1. **Charged cost.** Throughout this section, κ means the charged conceptual/desugared cost used by the strict selector, not the raw size of every compiled auxiliary clause. This is the cost notion fixed in **????** and exposed operationally in the strict engine description of **??**.
2. **Compiled payload.** Especially for the HIT steps, the compiled telescope may contain framework-derived eliminators or computation payload beyond the charged conceptual shell. Those extra compiled clauses can increase the evaluated novelty, but they do not increase the charged κ. This is the conceptual/compiled separation described in **??**.
3. **Uniqueness.** Every uniqueness claim below means "unique winner in the current strict unguided lane under admissibility, semantic minimality, and minimal overshoot". It does *not* assert a foundation-independent uniqueness theorem over all logics or all search policies.

Two further scope boundaries matter for the late stages. First, the codebase contains additional family-specific recovery and quality machinery in non-canonical lanes, but the present section does not use those heuristics as evidence for the strict claim. Whenever the verified strict trace already yields a singleton viable set, we use that stronger fact directly. Second, the proof-strength boundary from **??** is maintained verbatim: steps 1–10 admit exact or exact-reconstruction language, steps 11–14 are expository reconstructions constrained to match the strict totals, and step 15 canonizes only the strict total $\nu = 88$.

1.2 A Global Bar Lemma

Lemma 1.1 (Bar computation along the strict genesis trace)**.** *For every step $n \geq 2$ in the strict genesis trace,*

$$\mathrm{Bar}_n = \Phi_n \Omega_{n-1} = \frac{F_n}{F_{n-1}} \cdot \frac{\sum_{i=1}^{n-1} \nu_i}{\sum_{i=1}^{n-1} \kappa_i}.$$

Consequently the active bars for steps 2 through 15 are exactly the values listed in table 1.

Proof. By **??**, the current $d = 2$ lane satisfies $\Delta_n = F_n$, hence

$$\Phi_n = \Delta_n / \Delta_{n-1} = F_n / F_{n-1}.$$

By **??**,

$$\Omega_{n-1} = \frac{\sum_{i=1}^{n-1} \nu_i}{\sum_{i=1}^{n-1} \kappa_i}.$$

By **??**, the selection bar is $\mathrm{Bar}_n = \Phi_n \Omega_{n-1}$. Substituting the already realized strict-trace values (ν_i, κ_i) for $i < n$ therefore determines each bar uniquely. The worked arithmetic in **??** checks this explicitly for steps 5 and 6, and the same substitution yields the full bar column in table 1. □

Table 1: Strict genesis trace used in the proof. The 'cap/bands' column reports the strict admissibility cap together with the searched exact κ-band(s). The 'raw/dist.' column reports raw and deduplicated candidate counts. 'Runner-up' and 'near miss' mean the top displayed strict-lane competitor in the bar-clearing and subcritical pools respectively; a dash means that no distinct competitor was displayed by the verified strict trace.

Step	Structure	Δ_n	ν	κ	ρ	Bar	Margin	cap/bands	raw/dist.	Runner-up	Near miss
1	Universe	1	1	2	0.50	0.50	0.00	2 / 2	2144 / 552	—	—
2	Unit	1	1	1	1.00	0.50	0.50	1 / 1	96 / 78	—	—
3	Witness	2	2	1	2.00	1.33	0.67	1 / 1	96 / 81	—	found. cand. (1.00)
4	Π/Σ	3	5	3	1.67	1.50	0.17	3 / 3	1 / 1	—	—
5	S^1	5	7	3	2.33	2.14	0.19	3 / 3	1 / 1	—	—
6	PropTrunc	8	8	3	2.67	2.56	0.11	4 / 3	3 / 3	S^2 (3.33)	S^1 (2.33)
7	S^2	13	10	3	3.33	3.00	0.33	4 / 3	2 / 2	—	S^1 (2.33)
8	S^3	21	18	5	3.60	3.43	0.17	5 / 5	1 / 1	—	—
9	Hopf	34	17	4	4.25	4.01	0.24	5 / 4	1 / 1	—	—
10	Cohesion	55	19	4	4.75	4.46	0.29	5 / 4	2 / 2	—	Hopf (4.25)
11	Connections	89	27	5	5.40	4.91	0.49	6 / 5	2 / 2	—	S^3 (3.60)
12	Curvature	144	35	6	5.83	5.47	0.36	6 / 5|6	3 / 3	—	S^3 (3.60)
13	Metric	233	47	7	6.71	6.07	0.65	8 / 7	3 / 3	—	HIT (2.29)
14	Hilbert	377	63	9	7.00	6.78	0.22	9 / 8|9	4 / 4	—	HIT (3.00)
15	DCT	610	88	8	11.00	7.51	3.49	9 / 8	3 / 3	—	HIT (3.00)

1.3 Bootstrap: Steps 1–4

The bootstrap steps establish the minimum typed infrastructure from which later homotopical and geometric structure can be sealed. The foundational rule counts used here are the exact strict values recorded in **????** and table 1.

Proposition 1.2 (Step 1: Universe)**.** *At Step 1 the strict genesis trace selects the universe package.*

Proof. **What it is..** The selected structure is the first universe $\mathcal{U}_0$, namely the ambient type of small types. In the strict trace it is the unique selected bootstrap object at step 1; see table 1.

Why κ has this value.. The charged cost is $\kappa = 2$, exactly the two foundational clauses audited in **??**: universe formation and the universe-level witness. This is the minimal non-empty package that realizes a type universe rather than an opaque constant, so the cost is fixed by **??**.

How it builds on the established library.. It builds on nothing, because the prior library is empty. Precisely for that reason it is the only admissible way to create a typed growth regime at all: every later candidate presupposes some ambient type formation.

Why the novelty count has this value.. Its novelty is the exact foundational count $\nu = 1$ from **??**: one new formation rule for small types. There is no earlier eliminator to interact with, so no additional cross-interface novelty is available at this step.

Why it is the unique selected winner at this step.. The strict trace opens cap/band 2/2, evaluates 2144 raw candidates, and selects Universe with $\rho = 0.50$ exactly at the initial bar 0.50; see table 1. By the bootstrap admissibility conditions encoded in the strict search and by **??**, this is the unique least-supercritical bootstrap package. □

Proposition 1.3 (Step 2: Unit)**.** *At Step 2 the strict genesis trace selects the unit type* **1**.

Proof. **What it is..** The selected structure is the terminal one-point type **1**. It is the first concrete small type over $\mathcal{U}_0$, and therefore the first place where inhabitation can later be witnessed.

Why κ has this value.. The charged cost is $\kappa = 1$ because only the unit-type formation clause is irreducibly new at this step; see **??**. No extra eliminator clause is charged yet, since the selected package here is the type former itself, not its witness.

How it builds on the established library.. It leverages Step 1 exactly once: $\mathbf{1}$ is introduced as a small type in $\mathcal{U}_0$. This makes Step 2 the first genuinely inhabited target available for subsequent introduction rules.

Why the novelty count has this value.. The strict total is again exact and equals $\nu = 1$; see **??** and table 1. At this stage the unit type contributes one new foundational rule, but still no non-trivial interface interaction.

Why it is the unique selected winner at this step.. The strict trace opens cap/band 1/1 and selects Unit at $\rho = 1.00 > 0.50$, with no displayed runner-up or near miss; see table 1. Since the winner clears the bar with the smallest available cost and bootstrap admissibility still restricts the search to one-clause packages over the new universe, **??** forces this choice. □

Proposition 1.4 (Step 3: Witness)**.** *At Step 3 the strict genesis trace selects the unit witness* $\star : \mathbf{1}$.

Proof. **What it is..** The selected structure is the canonical inhabitant of the unit type. Semantically it is the first realized term, not merely the first realized type.

Why κ has this value.. The charged cost is $\kappa = 1$, because the selected specification adds exactly one irreducible introduction clause, namely the witness itself; see **??**. The associated elimination behavior is not charged separately at this step because it is accounted for via adjoint completion rather than as an additional primitive clause.

How it builds on the established library.. It uses the already selected unit type as its codomain and thus turns the bootstrap library from a pure type-formation fragment into an inhabited theory. This is exactly the bridge from Step 2's abstract type to a concrete term.

Why the novelty count has this value.. By **????**, the witness contributes one introduction rule and one elimination action, giving the exact strict total $\nu = 2$ recorded in table 1. This is the canonical early example where eliminative payload exceeds the raw charged clause count.

Why it is the unique selected winner at this step.. The strict trace opens cap/band 1/1 and selects Witness with $\rho = 2.00$ against bar 1.33; the displayed top near miss is only a foundational candidate at $\rho = 1.00$; see table 1. Hence **??** picks the witness package as the unique bar-clearer with minimal overshoot in the foundational one-clause regime. □

Proposition 1.5 (Step 4: Dependent function and sum types)**.** *At Step 4 the strict genesis trace selects the dependent function/sum package (Π/Σ types).*

Proof. **What it is..** The selected structure is the minimal former package supplying dependent functions, dependent sums, and the core application/pairing interface. Although the engine names the winner "Pi" in the trace, the mathematical structure is the joint Π/Σ former package described in **??**.

Why κ has this value.. The charged cost is $\kappa = 3$, corresponding to the selected specification listed in **??**: λ-introduction, pair formation, and application. This is the minimal cost at which the library acquires bona fide dependent former structure.

How it builds on the established library.. It builds on the previous three bootstrap entries by turning the bare inhabited universe into a theory with dependent abstraction and pairing. That infrastructure is the prerequisite for all later HIT, map, and API-shell steps.

Why the novelty count has this value.. By **??**, the exact foundational total is $\nu = 5$: two introduction rules (abstraction and pairing) together with three elimination rules (application, first projection, second projection). This is the strict total reported in table 1.

Why it is the unique selected winner at this step.. The strict trace opens cap/band 3/3 and finds a singleton viable set, with $\rho = 1.67$ against bar 1.50; see table 1. Since Step 4 is the first stage where the library needs a genuine former layer, and since the selected package is already constructively prime in the sense of **????**, it is uniquely selected. □

1.4 Geometric Ascent: Steps 5–9

The next five steps move from bootstrap syntax to genuine higher structure. Here the key theorem-level inputs are the two-layer coherence results **????**, the HIT/map exactness status from **??**, and the Agda abstraction-barrier evidence reported in **??**.

Proposition 1.6 (Step 5: The circle)**.** *At Step 5 the strict genesis trace selects the circle S^1.*

Proof. **What it is..** The selected structure is the first higher inductive type, the circle S^1, with a point and a non-trivial loop. This is the library's first genuinely homotopical object.

Why κ has this value.. The charged cost is $\kappa = 3$, because the selected conceptual shell is the minimal three-clause HIT signature from **??**: formation, base point, and loop. The compiled telescope is larger in the strict trace, but by the proof contract that extra compiled payload is not extra charged cost.

How it builds on the established library.. It is the first higher object that can be expressed only after the Π/Σ former layer has been installed. Thus it turns the bootstrap library into a setting where non-trivial path data can be generated and reused.

Why the novelty count has this value.. By **??**, the exact total is $\nu = 7$: five introduction-style schemas together with two computation/path rules coming from the loop constructor. This is the strict HIT reconstruction recorded in **??** and table 1.

Why it is the unique selected winner at this step.. The strict trace opens cap/band 3/3 and the viable set is singleton, with $\rho = 2.33$ against bar 2.14; see table 1. By theorem 1.1 and **??**, the circle is therefore the unique least-supercritical entry into the geometric regime. □

Proposition 1.7 (Step 6: Propositional truncation)**.** *At Step 6 the strict genesis trace selects propositional truncation.*

Proof. **What it is..** The selected structure is the propositional truncation operator $\|-\|_0$, which collapses higher structure to a mere proposition while still remembering existence. This is the first operation that systematically controls homotopical complexity rather than only creating it.

Why κ has this value.. The charged cost is again $\kappa = 3$, with the conceptual shell audited in **??** as truncation formation, truncation introduction, and the squash witness. As with S^1, the strict compiled telescope is larger than the charged conceptual shell, but the selector charges only the conceptual package.

How it builds on the established library.. It builds directly on the new geometric branch opened by S^1. In the library-growth dynamics this is the first point where PEN can both generate higher data and quotient it to mere existence, which is why the subsequent lift to S^2 becomes available.

Why the novelty count has this value.. The strict total is the exact HIT reconstruction $\nu = 8$ recorded in **??**. Operationally this counts the new truncation former, its introduction and elimination payload, and the squash-induced computation structure physically present in the strict HIT compilation.

Why it is the unique selected winner at this step.. This is the tightest selection point in the whole trajectory. The strict trace opens cap/band 4/3 and finds exactly two viable candidates: Trunc at $\rho = 8/3$ and S^2 at $\rho = 10/3$; see table 1. Since both have the same charged $\kappa = 3$, **??** reduces to minimal overshoot, and Trunc wins because its margin is only 0.11 whereas S^2 overshoots by about 0.77. The worked arithmetic in **??** shows that even one extra charged clause would kill this step, so the winner is both unique and structurally critical. □

Proposition 1.8 (Step 7: The 2-sphere)**.** *At Step 7 the strict genesis trace selects the sphere S^2.*

Proof. **What it is..** The selected structure is the first genuinely two-dimensional sphere, realized as a minimal higher inductive package. It is the next geometric lift after the circle/truncation bridge has been installed.

Why κ has this value.. The charged cost remains $\kappa = 3$, because the selected conceptual shell is the three-clause S^2 signature from **??**: formation, base point, and 2-path surface witness. Again the strict selector charges the conceptual shell, not every derived compiled clause.

How it builds on the established library.. By **????**, Step 7 inherits the two-layer interface debt created by Steps 5 and 6. The geometric meaning is exactly that S^2 is the first post-trunc higher lift: the library now has both non-trivial loop data and a way to compress it.

Why the novelty count has this value.. The strict total is the exact HIT reconstruction $\nu = 10$ listed in **??** and table 1. It is larger than the circle's total because a 2-dimensional HIT carries a richer path/computation payload in the strict evaluator.

Why it is the unique selected winner at this step.. The strict trace opens cap/band 4/3 and the viable set is singleton: S^2 clears the bar at $\rho = 3.33$, while the displayed near miss is only the replayed S^1 package at $\rho = 2.33$; see table 1. Thus **??** again yields a unique winner. □

Proposition 1.9 (Step 8: The H-space on S^3)**.** *At Step 8 the strict genesis trace selects the H-space package on S^3.*

Proof. **What it is..** The selected structure is not merely a bare 3-sphere, but the minimal multiplicative H-space package on S^3. This is the first point where PEN prefers a slightly richer geometric shell to a bare sphere because the additional interaction is worth the extra cost.

Why κ has this value.. The charged cost is $\kappa = 5$, exactly the H-space specification selected in **??**. The paper's comparison in **??** explains why this is the relevant charged shell: bare 3-sphere encodings exist at lower κ, but they overshoot the bar too much and are therefore not selected.

How it builds on the established library.. By **??**, Step 8 resolves the Fibonacci debt $21 = 13 + 8$ inherited from Steps 7 and 6. Geometrically, it uses the just-installed S^2 layer together with the truncation bridge to support the first multiplicative higher sphere.

Why the novelty count has this value.. The strict total is the exact HIT/H-space reconstruction $\nu = 18$ from **??**. More specifically, **??** shows that the bare 3-sphere already contributes 15 rules, and the H-space laws add 3 further elimination-style rules, yielding 18 in total.

Why it is the unique selected winner at this step.. The verified strict trace already has a singleton viable set at cap/band 5/5; see table 1. The paper's stronger specification comparison in **??** then explains the uniqueness semantically: among the natural S^3 presentations that clear the bar, minimal overshoot selects the H-space package rather than the bare suspension or the bare native HIT. □

Proposition 1.10 (Step 9: The Hopf fibration)**.** *At Step 9 the strict genesis trace selects the Hopf fibration.*

Proof. **What it is..** The selected structure is the map package $h : S^3 \to S^2$ classifying a principal S^1-bundle over S^2. This is the first step where the genesis sequence crosses from higher types to a non-trivial higher map.

Why κ has this value.. The charged cost is $\kappa = 4$, exactly the selected Hopf specification audited in **??**: map, fiber, total-space witness, and classifying map. Unlike the earlier HIT steps, the conceptual shell and the charged shell agree here.

How it builds on the established library.. It leverages the full preceding homotopy package: the domain S^3, the codomain S^2, and the fiber S^1 are all earlier library entries. By **??** and the Agda evidence summarized in **??**, the inherited obligation count is exactly $34 = 21 + 13$, with a domain/codomain split reflecting the S^3/S^2 interface.

Why the novelty count has this value.. For Step 9 the paper gives a sharp qualitative description: **??** records that $\nu_G = \nu_H = 0$ and $\nu = \nu_C$. Thus the exact strict total $\nu = 17$ from **??** and table 1 is entirely interactional map novelty rather than new type-former novelty.

Why it is the unique selected winner at this step.. The strict trace opens cap/band 5/4 and the viable set is singleton, so Hopf is already the unique strict bar-clearer at $\rho = 4.25$ against bar 4.01; see table 1. This is the point where the codebase's structural regime shifts from higher types to higher maps, but in the canonical strict lane no additional rescue heuristic is needed: the viable set has already collapsed to a single winner. □

1.5 Framework Abstraction: Steps 10–14

From Step 10 onward the sequence grows by library specifications rather than by new foundational judgment forms; see **??**. The semantic force of each step comes from constructive irreducibility and structural unity, not from opaque axiom packing. For Step 10 the rule audit is exact; for Steps 11–14 it is expository but constrained to match the strict totals by **??**.

Proposition 1.11 (Step 10: Cohesion)**.** *At Step 10 the strict genesis trace selects the cohesive modality package.*

Proof. **What it is..** The selected structure is the adjoint quadruple of modalities $(\flat, \sharp, \mathrm{Disc}, \Pi_\infty)$ described in **??**. This is the first abstraction step: PEN stops adding one more homotopy object and instead adds a modal interface for the whole geometric library.

Why κ has this value.. The charged cost is $\kappa = 4$, exactly the four modality-formation clauses listed in **????**. By **????**, these are irreducible API clauses: the adjoint string cannot be obtained definitionally from the prior HoTT library.

How it builds on the established library.. By **??**, a late API shell is valuable only insofar as it interacts with earlier structure. That interaction is explicit here: **??** records cross-interactions with S^1, S^2, S^3, and Hopf, so Cohesion is a global abstraction of the entire geometric ascent.

Why the novelty count has this value.. For Step 10 the rule audit is exact: **??** gives $\nu = 19 = 2 + 17$, namely two introduction-style unit maps together with seventeen elimination/interaction rules. This exact total is the one selected in the strict trace of table 1.

Why it is the unique selected winner at this step.. The strict trace opens cap/band 5/4 and again finds a singleton viable set: Cohesion clears the bar at $\rho = 4.75$, while the displayed near miss is the old Hopf package at $\rho = 4.25$; see table 1. Hence **??** selects the cohesive shell uniquely, and the non-survival of Hopf shows that PEN is now in a genuinely new abstraction regime. □

Proposition 1.12 (Step 11: Connections)**.** *At Step 11 the strict genesis trace selects the connection package.*

Proof. **What it is..** The selected structure is the differential-geometric package centered on a connection ∇ together with transport, covariant lift, horizontal lift, and Leibniz interaction; see **??**. This is the first differential shell built over the cohesive interface.

Why κ has this value.. The charged cost is $\kappa = 5$, exactly the five API clauses listed in **????**. By **??**, a connection is not derivable from cohesion alone, so the package must be charged explicitly.

How it builds on the established library.. It leverages Cohesion as its immediate carrier and thereby translates the modal library into differential transport data. This is the first late step whose entire meaning is cross-interface leverage: without the cohesive layer from Step 10 there is no differential package to select.

Why the novelty count has this value.. The expository rule audit in **??** reconstructs the strict total as $\nu = 27 = 2 + 25$, where the new interaction comes from cohesive cross-rules, function-space rules, and library interactions with the previously selected HITs and Hopf. By **??**, this reconstruction is subordinate to, but consistent with, the strict total in table 1.

Why it is the unique selected winner at this step.. The strict trace opens cap/band 6/5 and the viable set is singleton: Connections clears the bar at $\rho = 5.40$, while the displayed near miss is only the old S^3 package at $\rho = 3.60$; see table 1. Thus the strict lane does not need any external steering here: the first differential shell is already the unique bar-clearing continuation. □

Proposition 1.13 (Step 12: Curvature)**.** *At Step 12 the strict genesis trace selects the curvature package.*

Proof. **What it is..** The selected structure is the curvature layer $R = d\nabla + \nabla \wedge \nabla$ together with Bianchi, holonomy, Chern–Weil, and characteristic-class structure; see **??**. It is the first shell whose meaning is "second-order differential structure" rather than just differential transport.

Why κ has this value.. The charged cost is $\kappa = 6$, exactly the six irreducible clauses audited in **????**. These clauses are constructively irreducible because the elimination behavior they expose is not definitionally recoverable from a bare connection package; see **????**.

How it builds on the established library.. It leverages Connections as its immediate carrier and Cohesion as its ambient modal base. This is the canonical next abstraction after Step 11: once the library can differentiate, the next non-derivable interaction is precisely the failure of flatness.

Why the novelty count has this value.. The expository audit in **??** reconstructs the strict total as $\nu = 35 = 2 + 33$, with the extra weight coming from interactions among curvature, connection, and cohesive structure. By **??**, that reconstruction is explanatory rather than a second oracle, but it agrees with the canonical strict total in table 1.

Why it is the unique selected winner at this step.. The strict trace opens cap/bands 6/(5|6) and nevertheless yields a singleton viable set: Curvature clears the bar at $\rho = 5.83$, while the displayed near misses are far below the bar; see table 1. Thus the switch from differential to curvature structure is not a naming convention but a mechanically unique strict continuation. □

Proposition 1.14 (Step 13: Metric plus frame bundle)**.** *At Step 13 the strict genesis trace selects the metric package rather than a bare tensor or scalar-heavy detour.*

Proof. **What it is..** The selected structure is the metric-plus-frame-bundle package described in **????**: a symmetric bilinear form together with Levi-Civita connection, geodesic, volume, Hodge star, Laplacian, and Ricci structure. This is the first step where PEN selects an explicitly geometric API bundle rather than a single operator.

Why κ has this value.. The charged cost is $\kappa = 7$, exactly the seven clauses audited in **????**. By **??**, this package is charged as a single structural bundle: the bare metric tensor is too weak, whereas the coupled bundle is the smallest constructively prime package that can survive the Step 13 bar.

How it builds on the established library.. It leverages Curvature and Connections directly and uses the cohesive/differential layers indirectly through those references. The key structural point, emphasized in **??**, is that the metric package is selected because it unlocks the whole chain $g \to \nabla_g \to R \to \mathrm{Ric}$ inside the already accumulated differential library.

Why the novelty count has this value.. The expository reconstruction in **??** gives $\nu = 47 = 4 + 43$, where the dominant term is the bounded coupling with the curvature and connection layers. The same section explains why a bare metric tensor would have $\rho \leq 3.0$, far below the Step 13 bar, so the strict total in table 1 is genuinely bundle-driven rather than tensor-driven.

Why it is the unique selected winner at this step.. The strict trace opens cap/band 8/7 and the viable set is singleton: Metric clears the bar at $\rho = 6.71$, whereas the displayed near miss is only an unnamed HIT candidate at $\rho = 2.29$; see table 1. Hence the selected metric bundle is uniquely forced by the strict objective, and **??** explains why smaller fragments are excluded. □

Proposition 1.15 (Step 14: Hilbert functional)**.** *At Step 14 the strict genesis trace selects the Hilbert-functional package.*

Proof. **What it is..** The selected structure is the operator-analytic package described in **????**: inner product, completeness, orthogonal and spectral decomposition, C^*-algebra structure, cross-links to Metric, Curvature, and Connections, and a functional derivative. This is the analytic culmination of the framework-abstraction phase.

Why κ has this value.. The charged cost is $\kappa = 9$, exactly the nine clauses audited in **????**. By **??**, none of this operator-analytic structure is definitionally free from the preceding geometric library.

How it builds on the established library.. It leverages the three immediately relevant layers: Metric, Curvature, and Connections. This is a textbook PEN abstraction step: the new package is not selected for opaque existence but because it compresses and re-exposes the interaction of the whole geometric stack in analytic form.

Why the novelty count has this value.. The expository reconstruction in **??** gives $\nu = 63 = 5 + 58$, dominated by the coupling to Metric, Curvature, and Connections. The same section also shows why the margin is small: even two extra charged clauses would drop the package below the Step 14 bar.

Why it is the unique selected winner at this step.. The strict trace opens cap/bands 9/(8|9) and still yields a singleton viable set: Hilbert clears the bar at $\rho = 7.00$, while the displayed near miss is only an unnamed HIT candidate at $\rho = 3.00$; see table 1. Thus the operator-analytic layer is the unique strict continuation of the metric bundle, with no competing bar-clearer in the verified run. □

1.6 Terminal Synthesis: Step 15

Step 15 is special. The mathematical interpretation as DCT is theorem-level and semantic, but the only canonical quantitative claim is the strict total $\nu = 88$; see **????**.

Proposition 1.16 (Step 15: Temporal-cohesive synthesis)**.** *At Step 15 the strict genesis trace selects the temporal-cohesive shell whose semantic completion is read as DCT.*

Proof. **What it is..** The selected structure is the temporal-cohesive extension audited in **??**: next modality, eventually modality, the map $\bigcirc \to \Diamond$, and the explicit exchange laws between temporal and cohesive structure. Its semantic completion is the DCT reading established in **??**.

Why κ has this value.. The charged cost is $\kappa = 8$, exactly the eight temporal-cohesive clauses audited in **??**. As stressed in **????**, the infinitesimal reading is *not* charged as an extra primitive clause; it is derived semantically from the temporal-cohesive shell already selected.

How it builds on the established library.. It builds directly on Cohesion and sits atop the already selected Hilbert layer. This is the terminal synthesis step because it fuses the earlier spatial logic of Step 10 with temporal operators in a way that internalizes the library's own evolution; see **????**.

Why the novelty count has this value.. Only the strict total $\nu = 88$ is canonical here; no finer split is asserted in this section. By **????**, the strict engine computes that total from the conceptual temporal shell together with the explicit path-style witness payload preserved by compilation. This is exactly the number reported in table 1.

Why it is the unique selected winner at this step.. The strict trace opens cap/band 9/8 and the viable set is singleton: DCT clears the bar at $\rho = 11.00$ against bar 7.51, while the displayed near miss is only an unnamed HIT candidate at $\rho = 3.00$; see table 1. This is also the largest absolute overshoot in the entire sequence. Thus the temporal-cohesive shell is not merely admissible but uniquely dominant within the strict lane at the terminal step. □

1.7 Conclusion of the Step-by-Step Proof

Theorem 1.17 (The strict genesis sequence)**.** *In the current strict unguided PEN lane, the unique step-by-step sequence of selected structures through Step 15 is*

$$\begin{aligned}
&\mathcal{U}_0,\ \mathbf{1},\ \star,\ \Pi/\Sigma,\ S^1,\ \|{-}\|_0,\ S^2,\ S^3,\\
&\text{Hopf},\ \text{Cohesion},\ \text{Connections},\\
&\text{Curvature},\ \text{Metric},\ \text{Hilbert},\\
&\text{DCT}.
\end{aligned}$$

For each $1 \le n \le 15$, the selected structure has the corresponding values of Δ_n, ν_n, κ_n, ρ_n, and Bar_n *listed in table 1.*

Proof. Combine the global bar formula of theorem 1.1 with the fifteen step propositions theorems 1.2 to 1.16. Each step proposition identifies the selected structure, justifies its charged cost and novelty total at the appropriate proof-strength level, and verifies that it is the unique strict-lane winner at its step. □

Corollary 1.18 (No Step 16 continuation in the current strict lane)**.** *No Step 16 candidate clears the selection bar in the current strict unguided lane.*

Proof. By theorem 1.17, Step 15 is the selected temporal-cohesive shell with the canonical strict total $\nu = 88$. By the halting theorem **??**, every internal Step 16 continuation has zero marginal external novelty and every disconnected continuation fails the positive Step 16 bar. Hence no Step 16 candidate clears the bar. □